Tea Leaf Disease Classification Using CNN and Transfer Learning

CSC4093

Programming Assignment-02

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1. Introduction

Tea leaf diseases significantly affect agricultural productivity and quality. Early detection of diseases such as *algal leaf*, *brown blight*, and *white spot* is essential for effective crop management.

This study applies deep learning techniques using Convolutional Neural Networks (CNNs) with transfer learning. Two pre-trained models, VGG16 and ResNet50, are implemented and compared to classify tea leaf diseases based on image data.

2. Methodology

2.1 Dataset

The dataset contains three classes: Algal Leaf , Brown Blight, White Spot Each class includes approximately 100 images. The dataset was divided into 80% training and 20% validation.

2.2 Data Preprocessing

Images resized to 224 × 224 pixels , Preprocessing improves generalization and reduces overfitting. Data augmentation applied: Rotation , Zoom , Horizontal and vertical flipping

2.3 Model Development

Two transfer learning models were used:

VGG16

- Pre-trained on ImageNet , Frozen convolutional base
- Added layers: Global Average Pooling, Dense (128, ReLU), Dropout (0.5), Output layer (Softmax)

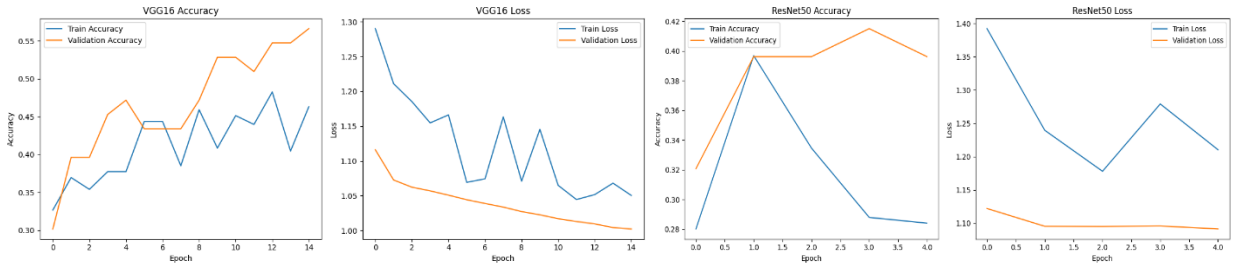
ResNet50

- Uses residual connections , Same classification head as VGG16, Frozen base layers

2.4 Training Setup

- Optimizer: Adam (learning rate = 0.0001)
- Loss function: Categorical Crossentropy
- Epochs: 15
- Early stopping applied to prevent overfitting

3. Results



The VGG16 model demonstrated better performance compared to ResNet50 in terms of both accuracy and loss. The ResNet50 model showed signs of underfitting, likely due to freezing all convolutional layers and limited dataset size. In contrast, VGG16 achieved more stable learning and better generalization, making it more suitable for this classification task.

	Model	Test Accuracy	Test Loss
0	VGG16	0.551724	0.999329
1	ResNet50	0.362069	1.122024

4. Conclusion

This project applied CNN-based transfer learning to classify tea leaf diseases into three categories.
Both VGG16 and ResNet50 models were evaluated using accuracy and loss metrics.
The results showed that **VGG16 performed better**, achieving higher accuracy and lower loss.
ResNet50 showed underfitting due to limited data and frozen layers.
Overall, VGG16 is more suitable for this task, while further improvements can be made through fine-tuning and larger datasets